Evidence-led concept report · 10 research papers

Personalised Learning System

A visual synthesis of how adaptive technology, learning science, learner psychology, and responsible AI can work together to improve learning efficiency.

Design for durable learning — not merely more screen time, more content, or faster answers.

Personalisation can help — when it is tied to learning quality

The literature points to positive potential, but effectiveness depends on what is being adapted, how feedback is delivered, and whether the system supports real understanding rather than passive engagement.

59%
Academic performance
In the 2024 higher-education scoping review, 41 of 69 studies reported improved academic performance.

36%
Engagement
25 of 69 studies in the same review reported increased learner engagement.

0.66 SD
Median ITS effect
A meta-analysis of 50 controlled evaluations found a median effect of 0.66 standard deviations for intelligent tutoring systems.

The strongest opportunity is not "AI content recommendation." It is adaptive support for what the learner needs to understand next.

RESEARCH COVERAGE



What the studies cover

System design
Learner profiles, recommendations, resource adaptation, and pathways.

Human behaviour
Goals, self-regulation, study behaviour, and learner differences.

Outcomes
Academic performance, engagement, and tutoring effectiveness.

From “what should I study?” to “what will help me learn?”

The system continuously learns from evidence of understanding, then chooses the next best learning action.

01
Diagnose
Map prior knowledge, misconceptions, goals, confidence, and time.

→

02
Plan
Build a flexible route around prerequisites and the learner’s objective.

→

03
Teach
Explain, question, demonstrate, and connect ideas in context.

→

04
Practise
Use retrieval, spacing, interleaving, feedback, and transfer.

→

05
Reflect
Review errors, confidence, strategy, retention, and independence.

Adaptive learner model

Track observable needs: prior knowledge, quiz performance, misconceptions, progress, goals, requests for help, confidence, and retention over time.

- Knowledge
- Behaviour
- Context

Evidence-based learning engine

Turn learning science into actions: retrieval practice, spaced practice, interleaving, elaboration, deliberate practice, scaffolding, and transfer.

- Memory
- Reasoning
- Application

A **AI tutor**
Explains, questions, demonstrates, and gives targeted feedback.

B **Learning coach**
Plans realistic sessions and develops self-regulated learning.

C **Attention layer**
Reduces distraction, interruptions, and unnecessary cognitive load.

What the system should do — and what it should avoid

The research suggests a human-centred approach: personalise based on evidence of need, while strengthening learner autonomy.

DO

Personalise observable needs

Use prior knowledge, performance, errors, goals, confidence, progress, and help-seeking behaviour.

AVOID

Fixed “learning styles”

Matching instruction to supposed visual, auditory, or kinaesthetic styles is not well supported by evidence.

DO

Make the learner retrieve

Ask for an attempt, explanation, prediction, or self-test before revealing the solution.

AVOID

Optimising screen time

Engagement metrics should not replace durable learning, transfer, and independent performance.

DO

Use AI as scaffolding

Provide hints, critique, questions, and feedback that gradually fade as competence increases.

AVOID

Letting AI do the thinking

Instant answers can produce shallow understanding and dependence if the learner never reasons first.

Core success measure

Can the learner remember, explain, apply, and transfer the knowledge later — with less support than before?

Responsible AI should preserve the learner’s thinking

The product can make AI assistance adjustable, transparent, and progressively less necessary.

Five assistance modes

1. **No-AI mode**
   Independent attempt before support.
2. **Hint mode**
   Prompts and partial cues, not solutions.
3. **Tutor mode**
   Step-by-step guidance with learner participation.
4. **Critique mode**
   Evaluate the learner’s own answer, reasoning, or work.
5. **Verification mode**
   Check evidence, sources, assumptions, and uncertainty.

Attention is a product feature

Use focused study sessions, one-task screens, friction around distractions, sensible breaks, low-distraction modes, and “resume learning” prompts.

Metacognition loop

Before: “What do I know?” During: “What is unclear?” After: “What mistake did I make, and what will I change?”

Plan — Choose a strategy → **Monitor** — Check understanding → **Adapt** — Change the approach

The system should become a better coach over time — and a smaller crutch.

Build a focused MVP around one learning journey

Start with one subject, one learner group, and one measurable outcome. Test whether the system improves durable learning rather than simply increasing activity.

1

Choose a domain

For example: university biology, mathematics, coding, language learning, or professional certification.

2

Map prerequisites

Represent the concepts, common misconceptions, and evidence needed to progress.

3

Measure learning

Compare delayed recall, transfer, confidence calibration, and independent performance.

Suggested MVP loop

Diagnostic quiz → learner model → adaptive micro-lesson → retrieval practice → feedback → spaced review → transfer task → reflection.

Evidence base used for this visual summary

[1] Xie et al. (2019), systematic review of technology-enhanced adaptive/personalised learning. *Computers & Education*.

[2] du Plooy et al. (2024), scoping review of personalised adaptive learning in higher education. *Heliyon*.

[3] Kulik & Fletcher (2016), meta-analysis of intelligent tutoring systems. *Review of Educational Research*.

[4] Gligorea et al. (2023), literature review of AI-based adaptive e-learning. *Education Sciences*.

[5] Kizilcec et al. (2017), self-regulated learning strategies in MOOCs. *Computers & Education*.

[6] Létourneau et al. (2025), systematic review of AI-driven intelligent tutoring systems in K–12 education. *npj Science of Learning*.

Note: Evidence figures and interpretations are reproduced from the supplied research summary. The source notes also caution against using fixed learning-style categories as the basis for personalisation.